

DECATUR, TX, USA

WMO: 720305

Lat: 33.254N

Lon: 97.581W

Elev: 319

StdP: 97.55

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53964

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.5	-3.2	-14.0	1.2	-1.2	-12.0	1.4	2.2	10.3	11.7	9.2	11.3	3.2	0	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.8	37.6	21.6	36.3	22.1	35.1	22.2	24.7	31.8	24.1	31.0	23.7	30.5	3.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	18.4	26.8	22.5	17.8	26.5	22.0	17.3	26.3	76.5	31.8	74.2	31.0	72.4	30.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.5	7.8	-9.3	39.1	2.5	1.4	-11.1	40.1	-12.5	40.9	-13.9	41.7	-15.7	42.7		
			-10.2	25.4	2.1	1.0	-11.8	26.1	-13.0	26.7	-14.2	27.2	-15.7	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	7.7	9.5	14.1	17.9	22.1	26.9	28.7	28.8	25.1	19.0	13.3	8.0
	DBStd	8.85	5.69	6.17	5.56	4.44	3.80	2.56	2.57	2.90	3.34	4.84	5.20	5.31
	HDD10.0	358	114	79	27	4	0	0	0	0	0	3	27	103
	HDD18.3	1344	332	252	151	59	13	0	0	0	1	52	163	321
	CDD10.0	3446	42	66	153	241	374	507	581	583	452	281	125	41
	CDD18.3	1388	1	4	18	46	129	257	322	325	203	71	11	1
	CDH23.3	15229	12	36	128	342	1102	2848	4018	4065	2026	577	68	7
	CDH26.7	7249	1	6	27	81	373	1334	2147	2213	886	170	12	0
Wind	WSAvg	3.7	3.7	4.0	4.5	4.5	4.1	3.9	3.4	3.1	3.0	3.5	3.7	3.6
Precipitation	PrecAvg	945	44	63	86	88	115	110	63	54	92	101	63	65
	PrecMax	1306	102	191	201	293	246	299	145	157	258	238	162	220
	PrecMin	616	2	1	2	9	18	31	1	0	17	13	5	3
	PrecStd	180	27	50	51	60	55	65	47	41	60	69	53	48
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.1	27.1	29.6	32.1	34.4	37.2	39.8	39.9	36.4	32.6	28.0	24.0
		MCWB	13.5	15.7	16.8	18.4	22.1	21.6	21.6	21.3	20.7	20.9	17.6	16.0
	2%	DB	22.0	23.8	26.9	28.8	32.3	35.3	37.5	37.8	34.6	30.5	25.1	21.0
		MCWB	12.3	14.9	16.8	18.5	21.5	21.9	21.6	21.7	21.7	20.3	16.4	13.9
	5%	DB	19.0	21.3	24.2	27.1	30.4	34.0	36.2	36.7	33.0	28.4	22.8	18.6
		MCWB	12.4	14.2	15.7	18.2	20.7	22.3	22.1	22.1	21.5	19.5	16.1	12.9
	10%	DB	17.1	19.0	22.4	25.1	28.9	32.7	34.8	35.2	31.6	26.9	21.1	16.3
		MCWB	11.3	13.5	15.1	17.4	20.6	22.0	22.3	22.4	21.1	18.2	15.2	11.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.9	19.0	19.9	21.8	24.3	25.6	25.4	25.2	24.6	23.1	20.8	18.3
		MCDB	19.3	23.0	25.5	26.4	30.7	32.4	32.8	32.9	30.6	27.9	24.9	20.9
	2%	WB	15.6	17.6	18.8	20.7	23.1	24.5	24.6	24.5	23.8	22.3	18.9	15.9
		MCDB	18.4	20.9	23.6	25.8	28.8	31.0	31.9	32.0	29.5	27.2	21.7	18.5
	5%	WB	13.7	16.1	17.8	19.9	22.3	23.8	24.0	23.9	23.2	21.5	17.7	14.3
		MCDB	17.9	19.5	21.6	24.5	27.8	30.0	31.3	31.3	28.7	26.2	20.9	16.9
	10%	WB	11.1	14.3	16.9	19.0	21.6	23.1	23.5	23.5	22.7	20.5	16.4	12.1
		MCDB	16.7	18.1	20.7	23.4	26.7	29.4	30.6	30.7	28.2	24.9	20.2	15.8
Mean Daily Temperature Range	5% DB	MDBR	10.7	10.8	11.1	11.1	10.2	10.4	10.7	10.8	10.5	11.0	10.7	10.1
		MCDBR	13.6	13.6	13.2	12.8	11.8	11.6	11.8	11.8	11.7	11.8	12.2	12.9
		MCWBR	7.9	7.6	6.3	6.3	4.5	3.2	3.0	3.0	3.4	5.1	6.4	7.7
	5% WB	MCDBR	12.1	12.3	10.5	10.7	10.1	10.4	10.5	10.4	9.7	10.4	10.3	10.9
		MCWBR	8.8	8.6	6.0	6.4	4.6	3.8	3.3	3.0	3.4	5.2	7.5	8.6
Clear-Sky Solar Irradiance	taub	0.303	0.314	0.346	0.378	0.414	0.427	0.429	0.425	0.389	0.350	0.324	0.308	
	taud	2.508	2.486	2.416	2.350	2.313	2.311	2.345	2.337	2.384	2.480	2.491	2.493	
	Ebn at Noon	918	941	931	910	875	859	856	855	874	889	884	889	
	Edh at Noon	84	96	111	124	130	130	125	124	114	95	86	81	
All-Sky Solar Radiation	RadAvg	2.92	3.48	4.55	5.53	5.95	6.68	6.77	6.21	5.17	4.04	3.10	2.55	
	RadStd	0.30	0.54	0.33	0.46	0.61	0.64	0.46	0.37	0.47	0.60	0.40	0.31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page